

Plan Proposal

Sprint 11 Project Plan Proposal

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Modifications



Modification #1: Enable Windows Sysmon and Defender in Wazuh with two custom local rules and a safe Active Response

On endpoint ad01, install Sysmon and ensure the channel **Microsoft-Windows-Sysmon, Operational** is collected by the Wazuh agent. Enable collection from **Microsoft-Windows-Defender, Operational**. Create two detections in `local_rules.xml`.

1. Alert on suspicious PowerShell use, for example `-EncodedCommand` or obvious base64 in the command line. This uses Sysmon Event ID 1 with the CommandLine field. ATT&CK, Command and Scripting Interpreter.
2. Alert when a local account is created or added to Administrators using Windows Security 4720 and 4732. ATT&CK, Account Creation and Account Manipulation.
Add a safe Active Response that only collects triage data on the host, for example process list, network sockets, recent event samples. Run it in audit mode first, then enable it. Capture screenshots of events, alerts, and the Active Response log.

References to use while implementing

- ☐ Wazuh documentation for Windows agent, Sysmon collection, Windows Defender collection, local rules, Active Response.
- ☐ Sysinternals Sysmon event reference, SwiftOnSecurity and Olaf Hartong configs.
- ☐ Microsoft Security auditing events 4720 and 4732.
- ☐ MITRE ATT&CK pages for T1059.001, T1136, T1098.
- ☐ TripleTen filesystem and network event PDFs.

Experiments



Experiment #1 Detect benign encoded PowerShell and verify Active Response

Generate a harmless encoded PowerShell command that prints a string, then execute it on ad01. Confirm Sysmon Event ID 1 with the full command line is ingested by Wazuh, the custom PowerShell rule alerts, and the triage Active Response runs. Record time from action to alert and list the triage fields used, user, host, parent process, full command line. Script Block Logging is optional and will be enabled only if available.

References to use while conducting

- ☐ Wazuh rule testing and Windows decoders.
- ☐ Sysmon Event IDs 1 and 7.
- ☐ PowerShell encoding basics.
- ☐ MITRE ATT&CK T1059.001.



Experiment #2 Detect privilege persistence, new local admin and scheduled task

Create a temporary local user, add it to Administrators, create a harmless scheduled task that echoes a string, then remove both. Verify Windows Security 4720, 4732, and 4698 appear in Wazuh and that the custom local admin rule alerts once. Note any tuning needed to avoid noise during maintenance. Record the most useful triage fields.

References to use while implementing

- ☐ Wazuh Windows Security event collection.
- ☐ Microsoft event IDs 4720, 4732, 4698.
- ☐ MITRE ATT&CK T1136 and T1053.

Experiment #3 Linux FIM, config tampering visibility on observer

Enable Wazuh File Integrity Monitoring on the Ubuntu observer host for `/etc` with `realtime yes` and `report_changes yes`. Make a reversible, harmless change in `sshd_config`, save, then restore the original. Also create and remove a simple test file in `/etc`. Verify alerts include path, user, old value, new value, and hashes. Contrast this filesystem signal with the Windows process signal from Experiment one.

References to use while implementing

- ☐ Wazuh File Integrity Monitoring and decoder documentation.
- ☐ OpenSSH `sshd_config` manual.
- ☐ ATT&CK context for configuration changes.
- ☐ TripleTen filesystem event PDFs.

Experiment #4 Detect risky AWS IAM changes with Wazuh CloudTrail ingestion

Enable CloudTrail to an S3 bucket and configure the Wazuh `aws-s3` module to ingest that bucket. In a lab account, perform `CreateAccessKey` for a test user and `AttachUserPolicy` with a broad policy. Confirm that Wazuh decoders parse the records and that a local rule alerts on these two actions. Capture time from action to alert, actor, target user, source IP, user agent, and AWS region..

References to use while implementing

- ☐ Wazuh `aws-s3` module and CloudTrail integration guide.
- ☐ AWS CloudTrail event name reference for `ConsoleLogin`, `CreateUser`, `CreateAccessKey`, `AttachUserPolicy`, `AssumeRole`.
- ☐ MITRE ATT&CK T1098 and T1078.